GEORGETOWN UNIVERSITY

DSAN 6725
Spring 2026

# DebateTrader: A Multi-Agent LLM Framework with Structured Bull–Bear Debate for Stock Trading

Multi-Agent · LLM Reasoning · Adversarial Debate · Portfolio Optimization · Backtesting

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| +41.0% | +64.3% |
|---|---|
| Total Return | Annualized Return |
| **3.052** | **72.2%** |
| Sharpe Ratio | Win Rate |

## 💡 MOTIVATION

Most LLM-based trading systems use a **single agent** to produce signals — prone to **confirmation bias** with no internal critique of conflicting evidence.

Single-agent bias · No internal debate · Limited interpretability

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**DebateTrader** decomposes each weekly decision into **analysis → adversarial debate → judgment → portfolio allocation**, making reasoning transparent and contestable at every stage.

## 📊 DATA SOURCES

| Source | Data Type | Freq. |
|---|---|---|
| **yfinance** | OHLCV prices | Daily |
| **SEC EDGAR** | 10-K / 10-Q | Quarterly |
| **FRED** | Macro indicators | Monthly |
| **Finnhub** | News headlines | Daily |
| **Google Trends** | Retail attention | Weekly |

Data filtered by **filing date** (not period-end) — prevents look-ahead bias.

## 🛡 RESPONSIBLE AI

- **Structured JSON outputs** across all stages — reduces hallucination surface
- **Rule-based factor extraction** before LLM synthesis — model sees only pre-extracted evidence
- **Deterministic fallbacks** at debate and portfolio stages — pipeline always produces a valid output even when LLM calls fail
- **Hard position & sector caps** in risk layer — LLM commentary cannot override allocation constraints
- **Paper-trading only** — no live broker execution; results are for research evaluation

No hallucinated data · Deterministic risk layer · Research prototype

**Limitations**

- Fixed 6-stock universe — not directly generalizable
- No transaction costs, slippage, or tax assumptions
- LLM priors on well-known companies may not be fully overridden by supplied evidence

## 🏗 SYSTEM ARCHITECTURE

**6 S&P 500 Stocks · 6 GICS Sectors**
AAPL (IT) · AMZN (Cons. Disc.) · BRK.B (Fin.) · GOOGL (Comm. Svcs.) · LLY (Health) · XOM (Energy)

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yfinance | SEC EDGAR | FRED | Finnhub | Google Trends

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ANALYST TEAM (parallel, per ticker)

- 📈 Technical — Rule + LLM
- 📊 Fundamental — Rule + LLM
- 🌍 Macro — Rule-only
- 📰 News & Trend — Rule-only

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DEBATE STAGE

🐂 Bull Agent ⚔ 🐻 Bear Agent

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⚖️ Judge Agent → Final Decision
Groq · Llama 3.1 8B · 3-week memory context

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🧠 Portfolio Judge — cross-ticker conviction weighting
Anthropic · Claude Sonnet 4.6 · single call for all 6 tickers

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🛡 Risk Management — position & sector caps, normalization

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📉 Backtesting — weekly rebalancing, 36 weeks (Aug 2025 – Apr 2026)

## 🤖 MODEL DESIGN

### 🐂🐻 Debate Stage

**Groq** · Llama 3.1 8B Instant

- Bull case from all 4 analyst reports
- Bear rebuttal with access to Bull's case
- Judge produces final signal & rationale
- Low temperature (0.2) + JSON-constrained prompting

### 🧠 Portfolio Judge

**Anthropic** · Claude Sonnet 4.6

- All 6 debate transcripts in one call
- Relative conviction-based weighting
- Forces cross-ticker comparison
- Fallback: confidence-weighted allocation

**Hybrid design:** Technical & Fundamental analysts use **rule extraction → LLM synthesis**. Macro & News analysts are **fully rule-based** (no LLM call). Risk management applies **deterministic constraints** before any LLM commentary is generated.

**Shared LLM Client**

Provider-agnostic wrapper supports both Anthropic and Groq. Selection via `DEBATETRADER_LLM_PROVIDER` env variable, with automatic fallback based on available API keys.

## 📈 BACKTEST RESULTS

36 weeks · Weekly rebalancing · Aug 2025 – Apr 2026

| Strategy | Return | Sharpe | Max DD | Win Rate |
|---|---|---|---|---|
| **DebateTrader** | **+41.00%** | **3.0521** | **9.72%** | **72.2%** |
| **Equal-Weight** | +29.01% | **3.0670** | 6.29% | 72.2% |
| **SPY B&H** | +7.86% | 1.1369 | 7.32% | 55.6% |
| **60/40** | +4.81% | 1.0892 | **4.94%** | 52.8% |

DebateTrader achieved the **highest raw return** (+41% vs +29% equal-weight) while maintaining a competitive Sharpe ratio. Max drawdown of 9.72% was higher than all benchmarks — a clear **return vs. risk trade-off**.

## 🧺 PORTFOLIO BEHAVIOR

- **36 unique allocation patterns** across 36 weekly decisions
- Active holdings per week: **3–6 tickers**, avg **4.53 per week**
- Portfolio judge produced **differentiated weights** — did not collapse to equal-weight
- 3-week debate memory provided **continuity** without overriding current-week analyst evidence
- Risk manager normalized 14 under-allocated portfolios (<100% total weight) across the backtest

## 🔍 KEY FINDINGS

- **Staged pipeline + JSON artifacts** made each component independently inspectable; intermediate outputs were reusable across debugging runs
- **Debate mechanism** surfaced competing interpretations — judge preserved dissenting points from the weaker side in most weekly decisions
- **Memory look-ahead risk** is subtle: temporal correctness must apply to agent state (memory artifacts), not only to market data
- **LLM reliability** was operationally significant — rate limits and parsing failures occurred during full backtest runs; deterministic fallbacks kept the pipeline running

**Future Directions**

Broader stock universe · Transaction costs & slippage · Debate quality metrics · Longer-horizon memory design

**DebateTrader** introduces structured adversarial debate into multi-agent LLM-based stock trading, combining deterministic feature extraction, constrained LLM reasoning, and rule-based risk management into a single interpretable pipeline. A 36-week backtest demonstrates competitive performance while surfacing important trade-offs in risk-adjusted returns and scalability.

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